Bioresource Technology Reports 34 (2026) 102779

Contents lists available at ScienceDirect

Bioresource Technology Reports

journal homepage: www.sciencedirect.com/journal/bioresource-technology-reports

# Optimizing $CO_2$ mitigation and biodiesel production using *Bacillus cereus* in bubble column bioreactor: A mathematical modeling approach for industrial upscaling

Rachael Jovita Barla [a,b], Suresh Gupta [a], Smita Raghuvanshi [a,*]

[a] Department of Chemical Engineering, Birla Institute of Technology and Science (BITS), Pilani, 333031, Rajasthan, India
[b] Australian Institute of Bioengineering and Nanotechnology, University of Queensland, St. Lucia, QLD, 4072, Australia

A R T I C L E I N F O

*Keywords:*
Chemolithotrophs
Biomass productivity
Carbon dioxide mitigation
Lipid productivity
Biodiesel yield
Fatty acid methyl esters analysis

A B S T R A C T

The utilization of *B. cereus* SSLMC2 in the bubble column bioreactor to facilitate $CO_2$ sequestration and biodiesel production is optimized by modeling the process for dissolved substrates, which predicts substrate uptake rate and biomass formation. The MATLAB *pdepe* solver was utilized to optimize the process parameters in the bioreactor. The highest biomass concentration was achieved with the upper range of parameters such as dissolved $CO_2$ and $O_2$, temperature, pressure, and gas flow rate. The lipid content and productivity of the biomass were found to be 56% and 0.1 g $L^{-1}$ $h^{-1}$, respectively. The transesterification process resulted in a biodiesel content and productivity of 89.3% and 0.2 g $L^{-1}$ $h^{-1}$, respectively. The evaluation of the biofuel based on the FAME profile also adhered to the established International standards for biodiesel. By adopting the approach above, it becomes feasible to effectively tackle the technological, energy, and environmental issues simultaneously.

## 1. Introduction

Global flue gas emissions from fossil fuels increased by 1% in 2023 compared to 2022, with projections indicating a further 0.3% rise in 2024 (UNEP Report, 2024). Achieving the emissions reduction targets of the Paris Agreement requires a 45% decrease in global greenhouse gas (GHG) emissions by 2030 (UN SDG Report, 2023). In this context, bacterial bio-mitigation of $CO_2$ offers a promising and sustainable approach for carbon capture and sequestration (CCS). Certain heterotrophic bacteria can perform chemolithotrophic metabolism, enabling the conversion of $CO_2$ into valuable biomolecules such as carbohydrates, lipids, and other organic compounds (Barla et al., 2024a, 2024b; Pekkoh et al., 2023). Compared with conventional industrial CCS technologies that often require high pressures and temperatures, bacterial bio-mitigation operates under ambient conditions, making it more energy-efficient and cost-effective. Moreover, integrating bacteria into bioreactor systems supports circular economy and waste-valorization principles, where captured $CO_2$ can be recycled into bio-based products such as biodiesel and bioethanol (Trapé et al., 2024). Although bioreactors can be designed in diverse sizes and configurations, their performance is commonly evaluated using empirical approaches based on correlations derived from experimental data under various operating conditions (Anand et al., 2023). However, this method provides limited mechanistic understanding of the process and often requires extensive experimentation to cover the full range of possible operating scenarios.

Utilizing mathematical modeling is beneficial for evaluating bacterial growth to control and forecast biomass production in a bioreactor. Furthermore, it assists in forecasting the most favorable operational parameters to enhance biomass productivity and facilitate $CO_2$ biofixation. The modeling approach aims to depict the bioreactor's actual and potential performance using established theory. This is achieved by expressing the theory mathematically, resulting in a functional process model (Dunn et al., 2013). Typically, mathematical models used to represent microbial systems consist of algebraic equations that describe microbial growth, nutrient absorption, and mass transfer, as well as nonlinear ordinary differential equations (Panda, 2011). Therefore, modeling compels a more comprehensive understanding of the subject, necessitating a critical evaluation of all relevant theories (Morchain, 2017). In order to optimize biomass production and $CO_2$ biofixation of *Chlorella vulgaris* in a continuous photobioreactor (PBR), Tebbani et al. (2014) employed a nonlinear predictive model. Empirical validation of the modeling results demonstrated that a $CO_2$ biofixation rate ranging

* Corresponding author at: Faculty Division-1, Department of Chemical Engineering, BITS PILANI, Pilani, 333031, Rajasthan, India.
*E-mail addresses:* p20190068@pilani.bits-pilani.ac.in (R.J. Barla), sureshg@pilani.bits-pilani.ac.in (S. Gupta), smita@pilani.bits-pilani.ac.in (S. Raghuvanshi).

https://doi.org/10.1016/j.biteb.2026.102779
Received 14 December 2025; Received in revised form 14 April 2026; Accepted 25 April 2026
Available online 2 May 2026

from 0.4 to 0.55 g $L^{-1}$ can be achieved with a 5% $CO_2$ concentration. The identical model was employed by Rodriguez-Jara et al. (2023) to optimize the growth of *Chlorella vulgaris*, resulting in a biomass production of 0.5 billion cells $L^{-1}$ at a dilution rate of 0.52 $h^{-1}$. Similarly, Al Ketife et al. (2016) constructed a mathematical model to evaluate the removal of nutrients and biofixation of $CO_2$, as well as to investigate the impact of nutrient concentration in a bioreactor. Anjos et al. (2013) employed the response surface approach to optimize the process of $CO_2$ bio-fixation by microalgae in a PBR. Multiple mathematical models exist in the literature for evaluating the development of microalgae in PBRs. Nevertheless, literature is scarce on the modeling of bacterial growth in bioreactors, particularly in bubble column reactors. Various parameters, including temperature, flow rate, $CO_2$ concentration, and nutrients, dictate the proliferation of bacteria. Thus, mathematical models are used to optimize these growth conditions. It can be employed for tasks such as process design, optimization, and control (Abu-Reesh, 2025). Inputting industrial or experimental data is necessary to establish or validate the model. However, the experimental data required is significantly less than for the empirical approach. In addition, the primary benefit acquired is the enhanced understanding of the process achieved solely by executing the modeling (Panda, 2011).

Modeling bacterial growth in a bioreactor and biodiesel extraction are interrelated components in bacterial biodiesel production systems. Modeling helps predict and optimize the growth phase during which maximum lipid accumulation occurs (Paleari, 2024). Innovative pathways are being explored to convert microbial biomass into biodiesel to harness the metabolic prowess of bacteria, offering a renewable and environmentally friendly alternative to traditional fossil fuels. Bacterial biomass serves as a promising feedstock for biodiesel production due to the ability of some bacteria to accumulate lipids, mainly triglycerides, within their cellular structure. These lipids can be extracted and transformed into biodiesel through a process known as transesterification (Ganta et al., 2019). Certain bacterial strains, such as species belonging to the genera *Rhodococcus*, *Pseudomonas*, and *Bacillus*, have demonstrated the ability to accumulate substantial amounts of lipids under specific growth conditions (Anand et al., 2023; Mishra et al., 2017). Various extraction methods, including solvent extraction and supercritical fluid extraction, are employed to recover lipids efficiently (Yadav et al., 2019). The choice of extraction method depends on factors such as cost, environmental impact, and scalability. The extracted lipids, primarily triglycerides, are subjected to transesterification, yielding biodiesel and glycerol as byproducts. The biodiesel produced from bacterial biomass is a mixture of fatty acid methyl or ethyl esters with properties comparable to traditional biodiesel derived from plant oils (Nayak et al., 2016). The production of biodiesel is completely determined by the magnitude of biomass production. A significant limitation encountered during industrial upscaling, $CO_2$ biofixation, and biodiesel production is the low biomass production (Abu-Reesh, 2024). Therefore, precise adjustment of the growing conditions in a bioreactor is an essential approach to improve the biomass yield in a bioreactor and the resulting biodiesel production.

Therefore, the novelty of this work lies in the development of an integrated mathematical model describing $CO_2$ mass transfer, bacterial growth kinetics, and biomass productivity in a bubble column bioreactor using *Bacillus cereus* SSLMC2. The model is implemented using the MATLAB PDEPE solver and validated using experimental data to identify optimal operating conditions for enhanced biomass and lipid production. The objectives of this study are to: (i) Develop a dynamic mathematical model describing $CO_2$ and $O_2$ mass transfer and bacterial growth in a bubble column bioreactor; (ii) Optimize key operational parameters affecting biomass production and $CO_2$ bio-mitigation; (iii) Evaluate lipid accumulation and biodiesel production potential from bacterial biomass; (iv) Provide predictive insights for industrial upscaling of bacterial $CO_2$ mitigation systems. This integrated modeling framework contributes to improving the design and optimization of bioreactor systems for sustainable biofuel production while simultaneously mitigating $CO_2$ emissions.

## 2. Materials and methods

### 2.1. Mathematical model for $CO_2$ bio-mitigation in the bubble column bioreactor

This study employs a dynamic model based on the overall material balance to comprehensively develop the process of $CO_2$ bio-mitigation and biomass production. Pioneering works by Luttmann et al. (1983) and Karl Schügerl (1982) contribute significantly to the theoretical framework of the model. These foundational studies provide essential insights into the intricacies of the processes involved, aiding in constructing a dynamic model that captures the essence of $CO_2$ mitigation and biomass production in bubble column reactors. The derivation of parameter values for the model is a critical step in ensuring its accuracy and applicability. The section of the bioreactor assumed for modeling purposes is represented in Fig. 1.

The model is built on presumptions that influence its formulation and application. These presumptions serve as the foundation upon which the entire theoretical framework rests. They represent simplified assumptions about the system that enable the development of a practical and workable model. While these presumptions may not capture all the nuances of the complex processes occurring within the bubble column reactor, they provide a valuable starting point for understanding and predicting system behavior.

i. The reactor consists of two phases, gas and liquid, with bacteria not considered as a distinct phase.
ii. The two-phase flow is characterized uniformly in both radial and angular directions. Plug flow is assumed for the bulk gas and liquid phases.
iii. The transport velocities and longitudinal dispersion coefficients remain consistent for all liquid phase components.
iv. Process variables exhibit angular and radial uniformity, excluding liquid velocity, with deviations evaluated solely along the longitudinal axis.
v. Mass transfer resistance is exclusively considered in the liquid phase. In the context of dissolved gases, dominance lies with the gas/liquid interface equilibrium.
vi. The model encompasses the material balance of specific components, such as biomass, and substrate ($CO_2$) in both the liquid and gas phases and $O_2$ in the gas and liquid phases.
vii. The model does not incorporate heat and momentum balances in its considerations.

A material balance on oxygen concentration in the liquid phase:

$$\frac{\partial C_L(x,t)}{\partial t} = D_L(t)\frac{\partial^2 C_L(x,t)}{\partial x^2} - u_L(t)\frac{\partial C_L(x,t)}{\partial x} - R_o(X,S,O,x,t) + k_L(x,t)\,a(x,t)\left[C_L^*(x,t) - C_L(x,t)\right] \tag{1}$$

A material balance on oxygen concentration in the gaseous phase:

$$p(x,t)\frac{\partial C_G(x,t)}{\partial t} = D_G(t)\frac{\partial}{\partial x}\left(p(x,t)\frac{\partial C_G(x,t)}{\partial x}\right) - \frac{\partial}{\partial x}\left(p(x,t)\,C_G(x,t)\,u_G(x,t) - \frac{R\,T\,\varepsilon_L(t)}{M_{O2}\,\varepsilon_G(t)}\right) + k_L(x,t)\,a(x,t)\left[C_L^*(x,t) - C_L(x,t)\right] \tag{2}$$

Initial conditions:

At the beginning of cultivation, before the medium is inoculated, the medium is saturated with $O_2$.

$$C_L(x,0) = C_L^*(x,0) \tag{3}$$

The inlet $O_2$ mole fraction in the gas is preserved along the column.

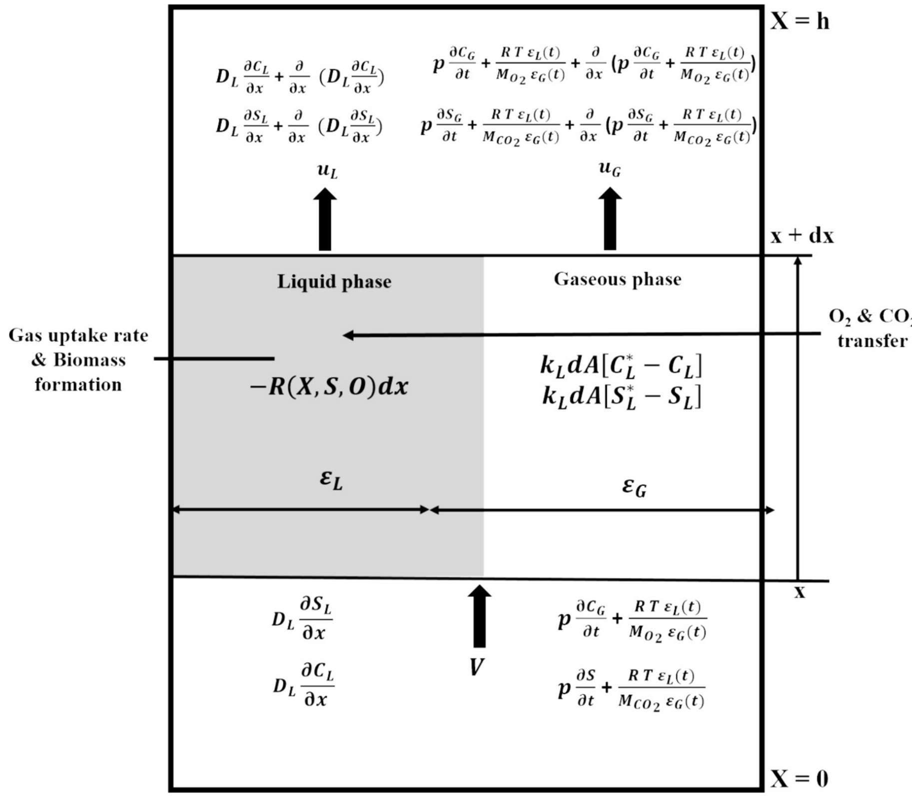


**Fig. 1.** The schematic view of the section of the bioreactor assumed for modeling purposes.

$$C_G(x,0) = C_{Go} \tag{4}$$

The boundary conditions for liquid and gas phases, respectively:

At the bottom boundary of the liquid

$$\frac{\partial C_L(0,t)}{\partial x} = \frac{u_L(t)}{D_L(t)}\left[C_L(0,t) - C_{Lo}(t)\right] \tag{5}$$

$$\frac{\partial C_G(0,t)}{\partial x} = \frac{u_G(t)}{D_G(t)}\left[C_G(0,t) - C_{Go}\right] \tag{6}$$

At the top boundary of the liquid

$$\frac{\partial C_L(h,t)}{\partial x} = 0 \tag{7}$$

$$\frac{\partial C_G(h,t)}{\partial x} = 0 \tag{8}$$

A material balance on the substrate ($CO_2$) concentration in the liquid phase:

$$\frac{\partial S_L(x,t)}{\partial t} = D_L(t)\frac{\partial^2 S_L(x,t)}{\partial x^2} - u_L(t)\frac{\partial S_L(x,t)}{\partial x} - R_S(X,S,O,x,t) + k_L(x,t)\,a(x,t)\left[S_L^*(x,t) - S_L(x,t)\right] \tag{9}$$

A material balance on $CO_2$ concentration in the gaseous phase:

$$p(x,t)\frac{\partial S_G(x,t)}{\partial t} == D_G(t)\frac{\partial}{\partial x}\left(p(x,t)\frac{\partial S_G(x,t)}{\partial x}\right) - \frac{\partial}{\partial x}\left(p(x,t)\,S_G(x,t)\,u_G(x,t) - \frac{R\,T\,\varepsilon_L(t)}{M_{CO2}\,\varepsilon_G(t)}\right) + k_L(x,t)\,a(x,t)\left[S_L^*(x,t) - S_L(x,t)\right] \tag{10}$$

Initial conditions:

At the beginning of cultivation, the substrate concentration is equal to the feed concentration.

$$S_L(x,0) = S_{Lo}(x,0) \tag{11}$$

The inlet $CO_2$ mole fraction in the gas is preserved along the column.

$$S_G(x,0) = S_{Go} \tag{12}$$

The boundary condition for liquid and gas phases, respectively:

At the bottom boundary of the liquid

$$\frac{\partial S_L(0,t)}{\partial x} = \frac{u_L(t)}{D_L(t)}\left[S_L(0,t) - S_{Lo}(t)\right] \tag{13}$$

$$\frac{\partial S_G(0,t)}{\partial x} = \frac{u_G(t)}{D_G(t)}\left[S_G(0,t) - S_{Go}\right] \tag{14}$$

At the top boundary of the liquid

$$\frac{\partial S_L(h,t)}{\partial x} = 0 \tag{15}$$

$$\frac{\partial S_G(h,t)}{\partial x} = 0 \tag{16}$$

The biomass balance in the column in the liquid phase:

$$\frac{\partial X_L(x,t)}{\partial t} = D_L(t)\frac{\partial^2 X_L(x,t)}{\partial x^2} - u_L(t)\frac{\partial X_L(x,t)}{\partial x} + R_X(X,S,O,x,t) \tag{17}$$

Initial condition:

$$X_L(x,0) = R_X(x,0) \tag{18}$$

The boundary condition for the liquid phase:

$$\frac{\partial X_L(0,t)}{\partial x} = \frac{u_L(t)}{D_L(t)}\left[X_L(0,t)\right] \tag{19}$$

$$\frac{\partial X_L(h,t)}{\partial x} = 0 \tag{20}$$

The governing equation for lipid accumulation in the column in the liquid phase:

$$\frac{\partial L(x,t)}{\partial t}=D_L(t)\frac{\partial^2 L(x,t)}{\partial x^2}-u_L(t)\frac{\partial L(x,t)}{\partial x}+R_X(X,S,O,x,t) \quad (21)$$

Initial condition:

$$L(x,0)=0\,(\text{no lipid initially}) \quad (22)$$

The boundary condition for lipid accumulation:

$$\frac{\partial L(0,t)}{\partial x}=\frac{u_L(t)}{D_L(t)}[L(0,t)] \quad (23)$$

$$\frac{\partial L(h,t)}{\partial x}=0 \quad (24)$$

The left-hand side (LHS) term in Eqs. 1 and 9 represents the rate of change of concentration of dissolved $O_2$ and $CO_2$, respectively, with respect to time. The term on the equation's right-hand side (RHS), $D_L(t)$, represents the liquid diffusion coefficient that varies with time. This coefficient quantifies the rate at which the $O_2$ and $CO_2$ diffuse in the spatial domain. The second spatial derivative of the concentration, referred to as the spatial curvature, quantifies the rate at which the concentration varies with spatial position. The second term, denoted as $u_L(t)$, represents the time-varying liquid flow velocity. The third term, $R_o$, describes the $O_2$ uptake rate that encompasses potential interactions or reactions among the species X, S, and O. The fourth term represents the chemical kinetics of substrate transfer across the concentration gradient in the system. In Eqs. 2 and 10, the LHS term represents the pressure variation that is correlated with both space and time, and it is related to the rate of change of the substance's mole fraction of $O_2$ and $CO_2$ with respect to time. The first term on the equation's RHS represents the combined effect of convection and diffusion in the x-direction, taking into account the pressure variation $p(x,t)$ and the gas diffusion coefficient ($D_G(t)$). The second term corresponds to the convective gas transport caused by the gas flow velocity $u_G(x,t)$ in the x-direction. The third term refers to the equation of state that describes the behavior of an ideal gas within the reactor. The fourth term bears a resemblance to the mass transfer rate in the preceding equation.

The LHS term of Eq. 17 represents the rate of change of biomass with respect to time. The first and second terms of the RHS represent parameters concerning the growth substrate ($CO_2$). The $R_X$ in the third term represents the correlation between dry-weight biomass and maximum specific growth rate. The above comprehensive equations encompass factors associated with diffusion, convection, chemical reactions, and processes that collectively elucidate the development of biomass, $O_2$, and $CO_2$ in a medium with spatial and temporal variations. The LHS term of Eq. 21 depicts the lipid concentration at position x along the reactor height and at time t. The terms on the RHS are similar to Eq. 17, except they correspond to the gradient of lipid concentration along the column height.

The modeling results in this paper have been obtained using ordinary differential equation (ODE) solver *pdepe* from ODEPACK in MATLAB R2024a. The present solver application is designed to solve 1-D parabolic and elliptic partial differential equations involving a single spatial variable $x$ and a time $t$. The solved equations are represented using *pdefun* notation, the initial value is represented using *icfun*, and the boundary conditions are represented using *bcfun*. The ODEs that arise from discretization in space are integrated to derive approximate solutions at the specified time intervals (*tspan*). It returns the values of the solution on a mesh given in *xmesh* using the *pdepe* function. Experiments conducted at varying $CO_2$ concentrations (g) in the previous study provide the necessary data to calibrate and fine-tune the model. These experiments offer a glimpse into the system's behavior under different conditions, allowing us to understand the interplay between $CO_2$ and $O_2$ concentrations, gas flow rates, diffusivity, solubility, and biomass production in the bioreactor.

## 2.2. Experimental parameters

The bio-mitigation experiments conducted in the previous work were subsequently utilized in this work for modeling and optimizing the process parameters (Barla et al., 2025). The biomass concentration, dissolved $CO_2$ (g), and dissolved $O_2$ (DO) were determined in the previous experimental study, and the values were incorporated in this study for modeling purposes (Barla et al., 2024a, 2024b). The specific growth rate (μ, $h^{-1}$), biomass productivity ($P_{max}$, g $L^{-1}$ $h^{-1}$), $CO_2$ bio-fixation rate (g $L^{-1}$ $h^{-1}$), and volumetric mass transfer coefficient values ($k_{La}$, $h^{-1}$) were also assumed based on the prior investigation (Barla et al., 2025). However, few calculations were made for the present study, utilizing values from the aforementioned prior studies.

The $O_2$ ($R_o$, g $L^{-1}$ $h^{-1}$) and $CO_2$ ($R_s$, g $L^{-1}$ $h^{-1}$) uptake rates and cell growth rate ($R_x$, g $L^{-1}$ $h^{-1}$) were estimated utilizing Eqs. 25 and 26 (Shuler et al., 2017).

$$R_o \text{ or } R_s=\frac{\mu X}{Y_{X/gas}} \quad (25)$$

$$R_x= \mu X \quad (26)$$

Where μ is the maximum specific growth rate ($h^{-1}$), X is the biomass concentration (g $L^{-1}$), and $Y_{X/gas}$ is the yield coefficient on $O_2$ or $CO_2$ (g cells/g $O_2$ or $CO_2$).

The gas ($\varepsilon_G$) and liquid ($\varepsilon_L$) hold-up in the media were calculated using Eqs. 27 and 28 (Shuler et al., 2017).

$$\epsilon_G=\frac{h-h_L}{h} \quad (27)$$

$$\epsilon_L=1-\epsilon_G \quad (28)$$

Where h is the height of liquid before aeration (m) and $h_L$ is the height of liquid after aeration in the reactor (m).

## 2.3. Biofuel production by transesterification

This study employed a binary solvent system for lipid extraction. Lipids were extracted using a chloroform-methanol combination in a 2:1 ratio. After extraction, the lipid was filtered, and its content was quantified gravimetrically after the evaporation of solvents detailed in the previous literature (Barla et al., 2025). The total lipid content (%) and productivity ($L_p$, g $L^{-1}$ $h^{-1}$) were calculated using Eqs. 29 and 30.

$$\text{Total lipid content}=\frac{\text{Weight of lipid (g)}}{\text{Weight of biomass (g)}}\times 100\% \quad (29)$$

$$L_p=\frac{\text{Total lipid content}}{100}\times P_{max} \quad (30)$$

Subsequently, 20 mL of methanol and 2% sulphuric acid ($H_2SO_4$) were mixed as catalysts with the lipids obtained. The mixture was thoroughly mixed and then reacted at 50 °C for 6 h in an incubator shaker. Centrifugation was carried out at 7000 rpm for 5 min to separate the organic and aqueous phase. To enhance biodiesel recovery from the organic phase, it was re-mixed in 5 mL of methanol and the mixture was vortexed and centrifuged at 3000 rpm for 5 min. The methanol portion was evaporated at 55 °C. The organic phase obtained after methanol removal was combined with 10 mL of n-hexane and washed thrice with 5 mL DI water. The hexane removal was achieved using the rotary evaporator. The GC–MS method was utilized to analyze the purity and fatty acid methyl esters (FAMEs) contents of biodiesel obtained from the in-situ transesterification reaction, following the details provided in the literature (Kumar and Thakur, 2018). The FAME profile was estimated by running it against standards prepared using the Supelco 37-component FAME mix (Sigma Aldrich, Germany). The yield of biodiesel (%)

and biodiesel productivity ($B_p$, g $L^{-1}$ $h^{-1}$) were calculated using Eqs. 31 and 32.

$$Biodiesel\ yield = \frac{Weight\ of\ biodiesel\ obtained\ (g)}{Weight\ of\ lipid\ (g)} \times 100\% \tag{31}$$

$$B_p = \frac{Biodiesel\ yield}{100} \times L_p \tag{32}$$

### 2.4. Estimation of biodiesel properties

The assessment of the biodiesel fuel quality was conducted to assess the quality of the bacterial biodiesel compared to other biodiesel and conventional diesel. It was examined by determining the FAME profile of *B. cereus* SSLMC2, following the empirical Eqs. 33 to 44 as outlined by (Nayak et al., 2018; Thangam et al., 2021). Properties of biodiesel, such as cetane number (CN), iodine value (IV, g $I_2$/g) saponification value (SV, g KOH/g), degree of unsaturation (DU, %wt), higher heating value (HV, MJ $kg^{-1}$), cloud point (CP, °C), specific gravity (SG, kg $L^{-1}$), viscosity (v, $mm^{-2}$ $s^{-1}$), density ($\rho$, g $cm^{-3}$), long chain saturated factor (LCSF, %wt), cold filter plugging point (CFP, °C) and oxidative stability (OS, h) were determined to evaluate the biodiesel quality. These fuel properties were also compared against ASTM D6751 standards to evaluate their suitability as biodiesel.

$$DU = \%MUFA + (2 \times \%PUFA) \tag{33}$$

$$IV = \sum \left( \frac{254 \times D_b \times \%F}{MW} \right) \tag{34}$$

$$CN = \left[ 46.3 + \left( \frac{5458}{SV} \right) \right] - (0.225 \times IV) \tag{35}$$

$$HV = 49.43 - (0.041 \text{x} SV) - (0.015 \times IV) \tag{36}$$

$$CP = (-13.356 \times DU) + 19.994 \tag{37}$$

$$SG = (0.0055 \times DU) + 0.8726 \tag{38}$$

$$SV = \sum \left( \frac{560 \text{x}\%F}{MW} \right) \tag{39}$$

$$v = -12.503 + 2.496 \times \ln\left( \sum MW \right) - 0.178 \times \sum D_b \tag{40}$$

$$\rho = 0.8463 + \left( 4.9 \Big/ \sum MW \right) + 0.0118 \times \sum D_b \tag{41}$$

$$LCSF = (0.1 \times C16) + (0.5 \times C17) + (1 \times C18) + (1.5 \times C20) + (2 \times C21) \tag{42}$$

$$CFP = (3.1417 \times LCSF) - 16.477 \tag{43}$$

$$OS = \frac{117.9295}{\%C18:2 + \%C18:3} + 2.5905 \tag{44}$$

where MUFA depicts the weight percentage of monounsaturated fatty acid, PUFA depicts the weight percentage of polyunsaturated fatty acid, $D_b$ represents the number of double bonds, F is the percentage of each fatty acid, and MW is the molecular weight of each fatty acid.

## 3. Results and discussion

The subsequent sections explore the findings derived from the validation study based on the experimental and modeling data of $CO_2$ biomitigation in a bubble column reactor. The collected data were subjected to statistical analysis using MATLAB R2024a software. The results also show the potential of biodiesel extraction from bacterial biomass.

### 3.1. Performance of experimental studies conducted under optimized conditions

When *B. cereus* SSLMC2 was cultivated using the optimized process parameters, it was observed that the growth rate and biomass productivity increased significantly. The highest biomass productivity was achieved when 25% $CO_2$ was used. These findings suggest that *B. cereus* SSLMC2 has the potential to be utilized for $CO_2$ fixation to enhance the production of biomass and lipids, which can be used for biodiesel purposes. Table 1 displays the various process parameters estimated during the previous experiments by Barla et al. (2024a, 2024b), utilized in model building at different concentrations of $CO_2$. These parameters obtained from the prior experiments are utilized further in the modeling. A brief explanation of the parameters obtained during the experimental studies is provided in supplementary section 1.1. The detailed experiments are present in the previous literature (Barla et al., 2025).

### 3.2. Prediction of biomass growth and dissolved $O_2$ and $CO_2$ by modeling equations

A simplified form of reaction taking place in a bioreactor given by Rittmann and McCarty (2001) is represented in Eq. 45.

$$Reactants + Cells\ (inoculum) \rightarrow Cell\ production + Products \tag{45}$$

The inoculum refers to the smallest quantity of viable cells introduced into the reactor to initiate the reaction. The reactants comprise carbon and nitrogen sources, $O_2$, minerals, and micronutrients. The reaction proceeds through the proliferation of cells in the inoculum, which consume the reactants and generate additional cells and products. Several variables, including the composition of the reactant media, the feed strategy, the condition of the inoculum, the pH level, temperature, and the shear forces, impact this reaction (Onyeaka and Ekwebelem, 2023). The sole substrate for growth in this study is $CO_2$, and $O_2$ is required as the reaction requires aerobic conditions. The model estimated the concentration of the dissolved gases in the liquid phase and their mole fraction in the gaseous phase. Among the four experiments, the cultivation with 25% $CO_2$ produced the most favorable outcomes. Therefore, the modeling results are exclusively based on the experimental results obtained with 25% $CO_2$. The results of modeling studies for 10, 15, and 20% $CO_2$ are presented in the supplementary section. The model predicts the concentrations of dissolved $CO_2$ and $O_2$ and biomass over time, as shown in Figs. 2 (a, c, and e), respectively. Fig. 2 (b, d, and f) shows a comparison between the experimental and model data collected for various concentrations of $CO_2$, $O_2$, and biomass, respectively.

The mesh diagram in Fig. 2 (a) illustrates the variation in the concentration of dissolved $CO_2$ in the liquid phase along the length of the column. The gradient from blue to yellow signifies the ascending sequence of concentration within the medium. As per the model, the $CO_2$ concentration in the medium gradually increases over time and remains relatively consistent throughout the reactor's length. The solubility of $CO_2$ is contingent upon factors such as pressure, the composition of the medium, temperature, and superficial gas velocity (Gevantman, 2015). According to Henry's law, higher pressure enhances the solubility of $CO_2$ in the brine solution (Zhou et al., 2012). As the temperature of the brine solution increases, its capacity to retain $CO_2$ diminishes, potentially resulting in the release of $CO_2$ as a gas (Gaur, 2015). Hence, the optimum temperature of 32 °C was maintained throughout all experiments to facilitate greater solubility. The solubility of $CO_2$ can also be influenced by the composition of the brine, particularly the presence of other ions or chemicals. The salinity of the brine can enhance the ability of $CO_2$ to dissolve by its interactions with the ions found in the saline solution (Setia et al., 2011). The pH of the brine can impact the generation of carbonic acid ($H_2CO_3$) due to $CO_2$ dissolution, thereby influencing the

**Table 1**
The process parameters obtained during the cultivation of *B. cereus* SSLMC2 from the previous experimental study.

| | $CO_2$ | | | | | | $O_2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| $CO_2$ (%) | Bio-fixation rate ($g\ L^{-1}\ h^{-1}$) | $P_{max}$ ($g\ L^{-1}\ h^{-1}$) | μ ($h^{-1}$) | $R_s$ ($\times 10^{-5}$ $g\ L^{-1}\ h^{-1}$) | $S_L$* ($g\ L^{-1}$) | $k_{La}$ ($h^{-1}$) | $R_o$ ($\times 10^{-5}$ $g\ L^{-1}\ h^{-1}$) | $C_L$* ($g\ L^{-1}$) | $k_{La}$ ($h^{-1}$) |
| 10 | 0.049 | 0.042 | 0.009 | 8 | 0.021 | 0.074 | 6 | 0.025 | 0.057 |
| 15 | 0.048 | 0.035 | 0.028 | 7 | 0.0062 | 0.055 | 3 | 0.0189 | 0.033 |
| 20 | 0.063 | 0.032 | 0.014 | 5 | 0.0125 | 0.074 | 2 | 0.0171 | 0.036 |
| 25 | 0.083 | 0.051 | 0.008 | 6 | 0.016 | 0.083 | 2.5 | 0.021 | 0.082 |

overall solubility (Mishra et al., 2018). The observed increase in solubility is consistent with the maintenance of optimal conditions in the reactor, which promotes the effective dissolution of $CO_2$.

Fig. 2 (b) shows that the experiment yielded a maximum dissolved $CO_2$ concentration of 0.028 $g\ L^{-1}$, while the model predicted a value of 0.033 $g\ L^{-1}$. The two values show a deviation of 16.2%, with the model predicting a concentration higher than the experimental value measured. The correlation coefficient ($R^2$) between the predicted and experimental values were 0.99, 0.82, 0.91, and 0.93 for 10, 15, 20, and 25% $CO_2$, respectively, giving the best fit for 10% $CO_2$. A deviation of 12.5%, 11.5%, and 12.5% is noted between the observed and modeled values of dissolved $CO_2$ for 15%, 20%, and 25% $CO_2$, respectively. In

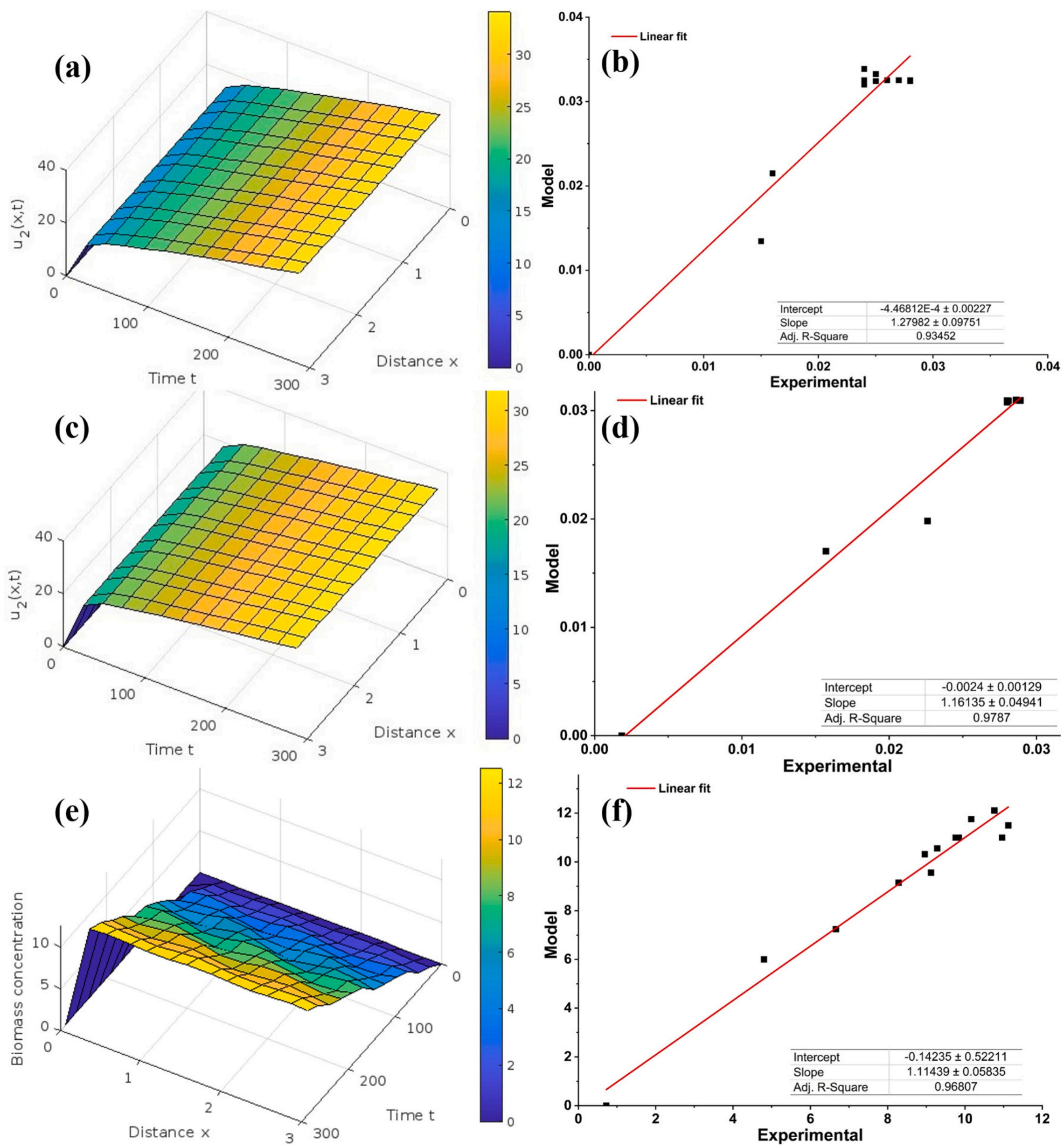


**Fig. 2.** The parameter model fitting obtained during 25% bio-mitigation experiments **(a)** dissolved $CO_2$, **(b)** dissolved $CO_2$ correlation between the experimental and model values, **(c)** dissolved $O_2$, **(d)** dissolved $O_2$ correlation between the experimental and model values, **(e)** biomass concentration, and **(d)** biomass concentration correlation between the experimental and model values.

each instance, the model's prediction exceeded the measured outcome. The variation may be due to factors such as measurement errors, model inaccuracies, or assumptions in the modeling process. It is crucial to consider the context of the data and the specific characteristics of the experiment to understand the importance of this difference.

The solubility of $O_2$ in a brine solution (Fig. 2. (c)) is affected by various factors analogous to the solubility of $CO_2$. Elevated pressure conditions generally result in increased $O_2$ solubility in brine. Unlike the solubility patterns of numerous gases, the solubility of $O_2$ in water generally diminishes as the temperature rises. With an increase in water temperature, the ability of $O_2$ to dissolve decreases, leading to its potential release into the atmosphere (Bok et al., 2023). The solubility of $O_2$ in brine can be significantly influenced by biological processes, such as the metabolism of microorganisms, thereby impacting the overall $O_2$ levels in the brine. Molecular $O_2$ acts as the electron acceptor in the aerobic breakdown of carbon compounds (Rittmann and McCarty, 2001).

Fig. 2 (d) compares the empirical and theoretical data for dissolved $O_2$ in 25% $CO_2$ cultivation media. The $CO_2$ input varied in each experiment, while the $O_2$ input remained consistent at a concentration of 20% in all trials. Fig. 2 (e) indicates that the experiment resulted in a peak dissolved $O_2$ concentration of 0.029 g $L^{-1}$, whereas the model projected a value of 0.031 g $L^{-1}$. The two values exhibit a deviation of 6.4%, with the model forecasting a concentration more significant than the actual experimental measurement. There is a discrepancy of 11.9%, 6.7%, and 8.5% between the measured and predicted values of dissolved $O_2$ for the experiment of 10%, 15%, and 20% $CO_2$, respectively. The $R^2$ between the predicted and experimental values of dissolved $O_2$ were 0.93, 0.98, 0.98, and 0.98 for 10, 15, 20 and 25% $CO_2$, respectively. The values fit each experiment well, projecting accurate dissolved $O_2$. Dissolved $O_2$ is consistently reduced in the sample because of oxidation of the inorganic compound or consumption by the bacteria to produce energy. The inconsistency between the measured and model values could result from fluctuations caused by the microorganism's respiration activity during measurement and contact with air at the air-water interface (Emerson et al., 2019).

Fig. 2 (e) illustrates the biomass formation prediction in the reactor in the presence of $CO_2$ and $O_2$. The projected biomass in all the experiments demonstrates a positive correlation between biomass concentration and column height. It undergoes exponential growth at first and then reaches a stable state over time. Bacteria utilize carbon as a fundamental component for their biomass build-up. Aerobic bacteria are capable of using $O_2$ for energy production through cellular respiration. This process entails the decomposition of organic substances obtained from the absorption of $CO_2$. $O_2$ acts as an electron acceptor in the electron transport chain, enabling the production of adenosine triphosphate (ATP), the cellular energy currency of microorganisms. Bacteria utilize $CO_2$ and $O_2$ in their metabolic activities to produce organic compounds such as proteins, lipids, and nucleic acids (Mishra et al., 2018). As biomass builds, bacteria undergo mitotic division, resulting in a proliferation of their population within the reactor. The rapid increase in biomass is a crucial element of its production and is affected by factors such as the availability of nutrients, environmental conditions (such as temperature and pH), and the concentration of gases like $CO_2$ and $O_2$ (Anand et al., 2020). The process of biomass synthesis in a reactor, which involves the interaction of $CO_2$ and $O_2$, is a dynamic phenomenon demonstrating the intricate relationship between microbial metabolism, carbon fixation, energy production, and cell growth. Comprehending and managing these parameters are crucial for enhancing biotechnological processes focused on generating biomass for diverse purposes.

Fig. 2 (f) illustrates the comparison between the observed biomass concentration and the model predictions for biomass growth under varying $CO_2$ levels. The best fit for the biomass concentration was obtained in the case of 25% $CO_2$. Fig. 2 (f) shows that the experiment yielded the highest biomass concentration of 11.12 g $L^{-1}$, while the model predicted a value of 12.1 g $L^{-1}$. The $R^2$ values obtained were 0.91, 0.86, 0.71, and 0.97 for 10, 15, 20 and 25% $CO_2$, respectively. The two values show a deviation of 8.1%, with the model predicting a concentration higher than the actual experimental measurement. There are 11.1%, 8.5%, and 12% discrepancies between the measured and predicted values of biomass concentration for the experiments with 10%, 15%, and 20% $CO_2$, respectively. The error could be caused by inaccuracies in measuring the dry weight of the sample or by withdrawing samples from the reactor.

Fig. S3 denotes the mole fraction of $CO_2$ at different concentrations of feed $CO_2$. The mole fraction of $CO_2$ in a brine solution quantifies the concentration of $CO_2$ with the total number of moles of all constituents in the mixture. The dimensionless quantity offers valuable information about the abundance of $CO_2$ in the brine. The mole fraction of $CO_2$ in the media is affected similarly to the dissolved $CO_2$. The mole fraction of $CO_2$ in the bioreactor plays a vital role in optimizing the bacteria's overall metabolism. Fig. 5 illustrates the mole fraction of $CO_2$ in the media. It demonstrates that the mole fraction is initially elevated in the media due to the introduction of $CO_2$ gas. Over time, it reaches a stable level in the media, ensuring an adequate supply of $CO_2$ as a carbon substrate to facilitate bacterial growth. Fig. S6 denotes the mole fraction of $O_2$ at different concentrations of feed $CO_2$. $CO_2$ and $O_2$ are the two crucial variables in the media that significantly impact the overall process. Chemical reactions in the media can influence the mole fraction of gases. The mole fraction of $O_2$ is a vital factor in evaluating oxygen availability when studying the survival of bacterial life (Mishra et al., 2019). The mole fraction of $O_2$ in the medium exhibited a comparable behavior to that of $CO_2$. The bacteria's continuous utilization of $CO_2$ and $O_2$ in the media resulted in a depletion of resources. However, due to the uninterrupted provision of the gaseous mixture in the reactor, the concentration of the gases was regulated within an optimal range to promote the thriving of the bacteria. The presence of $CO_2$ and $O_2$ in the media directly influences biomass formation. The growth would be limited if insufficient substrates were present in the media.

The experiment and model showed the best fit for 25% $CO_2$ with a mean squared error of 0.35. This shows a positive correlation between the experimental and predicted values without data overfitting. Based on existing literature, the error percentage calculated between the two values fell within an acceptable range (Nayak et al., 2018). The validation confirmed the efficacy of the model and reactor design. Additionally, the modeling approach led to a notable 12% increase in biomass productivity. Hence, the model successfully projected optimized $CO_2$ and $O_2$ concentration in the reactor, increasing biomass concentration due to sufficient growth factors. Reactor scale-up is transitioning a chemical or biological procedure from a smaller laboratory or pilot-scale to a larger full-scale industrial production. Modeling allows engineers to forecast and comprehend the reactor system's behavior in various scenarios, aiding in designing and enhancing the larger-scale production process. Models serve as a foundation for creating validation experiments on a larger scale to verify the precision of forecasts. Utilizing validated models can help reduce industrial scale-up risks by ensuring a thorough understanding of the reactor's behavior. Modeling values are also essential in the industrial scaling of reactors to get the desired product formation. Henceforth, the product formation potential of the biomass concentration acquired during the experiment will be examined in the subsequent section.

### 3.3. Variation of different process parameters to assess the model

To assess the accuracy of the model, the process parameters were systematically adjusted by a range of ±50% to evaluate the reliability of the model. The important parameters affecting the process are diffusivity, gas flow rate, temperature, and pressure. Fig. 3 (a) shows the dissolved $CO_2$ concentration changes in the reactor over time and distance, with a 50% increase in diffusivity of $CO_2$. Increased diffusivity leads to more uniform $CO_2$ dispersion in the reactor. The uniform yellow color along the x-axis shows efficient $CO_2$ dispersion in the reactor. Fig. 3 (b) shows biomass concentration with increased $CO_2$ diffusivity under the same conditions. Enhanced $CO_2$ diffusivity promotes higher biomass concentrations. Though there is more $CO_2$ present, the reactor's biomass distribution is slightly uneven, suggesting that other factors like flow rate, nutrients, or mixing may be affecting it (Alshalif et al., 2022). In Fig. 3 (c), dissolved $CO_2$ concentration is shown when $CO_2$ diffusivity is reduced by 50%. The concentration of $CO_2$ decreases with distance from the source, as shown by the color change from yellow to blue. This suggests ineffective $CO_2$ diffusion, leading to lower concentrations in specific reactor areas. Fig. 3 (d) shows biomass concentration with decreased $CO_2$ diffusivity. Lower $CO_2$ levels due to reduced diffusivity result in lower biomass levels, particularly in areas farther from the source (Mishra et al., 2019). The biomass distribution is more uneven than in Fig. 3 (b), indicating a greater impact of limited $CO_2$ availability. Similarly, a higher $CO_2$ flow rate results in more uniform dissolved $CO_2$ concentrations, promoting more evenly distributed biomass in the reactor (Fig. 3 (f)). A consistent yellow color indicates a higher concentration of dissolved $CO_2$ across the reactor, as shown in Fig. 3 (e). This suggests that $CO_2$ is more efficiently delivered and easily absorbed by organisms. The blue color gradient in the reactor indicates reduced biomass growth due to limited $CO_2$ availability because of the reduced gas flow rate. Hence, increased $CO_2$ diffusion and gas flow rate promote biomass growth but with some irregular dispersion. Reduced $CO_2$ diffusion and gas flow rate leads to decreased biomass, especially in areas far from the source.

A higher temperature results in a more uniform mole fraction of $CO_2$, promoting a more evenly distributed biomass concentration in the reactor. A consistent yellow color indicates higher $CO_2$ mole fractions throughout the reactor, indicating the positive impact of higher temperatures on $CO_2$ availability, as depicted in Fig. 4 (a). Higher biomass concentrations are evenly distributed across the reactor, as shown by the color gradient in Fig. 4. (b). The reactor's far end may have a slight biomass concentration drop due to other limiting factors. Lower temperatures may decrease $CO_2$ solubility and diffusion, resulting in a less uniform mole fraction across the reactor. Lower temperatures slow metabolic and growth processes, lowering biomass concentrations (Anand et al., 2024). Hence, lower temperature results in reduced $CO_2$ mole fractions which in turn results in lower biomass concentration, especially in areas further from the source (Figs. 4 (c & d)). Elevated pressure results in an elevated initial $CO_2$ mole fraction and biomass concentration (Figs. 4 (e & f)). Both parameters exhibit a more distinct increase as time and distance increase, suggesting a faster reaction or consumption. When the pressure is reduced, the outcomes yield reduced initial values for both the mole fraction of $CO_2$ and the concentration of biomass (Figs. 4 (g & h)). The decline is characterized by a more gradual rate, indicating a slower response and reduced production of biomass (Viswanaathan et al., 2022). Therefore, elevated temperature and pressure facilitate the proliferation of biomass, and decreased temperature and pressure levels result in a reduction in biomass, particularly in areas distant from the inlet source in the reactor.

Fig. 5 (a) illustrates the relationship between the concentration of dissolved $CO_2$ and the corresponding increase in pressure within the system. The correlation seems to be positive and approximately linear, suggesting that higher pressure (2 atm)leads to increased solubility of $CO_2$ (0.033 g $L^{-1}$) within the reactor. This may be attributed to an elevation in the partial pressure of $CO_2$, resulting in a greater amount of $CO_2$ being dissolved in the medium (Doucha et al., 2005). Fig. 5 (b) illustrates the correlation between temperature and dissolved $CO_2$. The temperature rise (315 K) is directly proportional to the increase in dissolved $CO_2$ concentration (0.032 g $L^{-1}$), indicating either an exothermic process or a system in which higher $CO_2$ solubility leads to enhanced metabolic activity, consequently producing heat (Ganta et al., 2019). The diffusivity of the medium is positively correlated with biomass concentration, suggesting that higher diffusion of gas promotes more biomass formation (Fig. 5 (c)). The highest biomass concentration of 14.2 g $L^{-1}$ was achieved when the diffusivity was increased up to $3 \times 10^{-5}$ cm $s^{-1}$. This may be due to the physical changes in the media or increased microbial activity blending. The gas flow rate also positively correlates with biomass concentration, indicating that a higher gas flow rate of 4 L $min^{-1}$ necessitates greater biomass levels up to 12.42 g $L^{-1}$, likely due to an elevated metabolic demand for $O_2$ and the mitigation of $CO_2$ (Fig. 5 (d)). The biomass concentration increases significantly as the rate of $CO_2$ uptake increases up to $8 \times 10^{-5}$ g $L^{-1}$ $h^{-1}$, suggesting that a higher amount of biomass actively consumes more $CO_2$ for growth or metabolic processes (Fig. 5 (e)). The rise in $CO_2$ uptake rate may be attributed to elevated respiratory activity within the system. Likewise, the concentration of biomass increases as the $O_2$ uptake rate increases up to $3.5 \times 10^{-5}$ g $L^{-1}$ $h^{-1}$ (Fig. 5 (f)). The observed pattern indicates that increased biomass levels lead to augmented $O_2$ consumption, most likely due to respiration and other oxidative activities occurring within the reactor. Therefore, both pressure and temperature exert a substantial influence on the concentration of dissolved $CO_2$ in the system, increasing dissolved $CO_2$ as both parameters increase. When the system demonstrates elevated diffusivity, increased gas exchange, and higher rates of both $CO_2$ and $O_2$ uptake, the concentration of biomass rises, indicating a boost in metabolic activity (Eloka-Eboka and Inambao, 2017).

### 3.4. Lipid production and fatty acid profile analysis

The lipid content and lipid productivity of bacterial biomass have garnered significant interest due to their direct correlation with the production of biodiesel. The performance of the increasing $CO_2$ concentration in the system was compared in terms of lipid and biodiesel production from the cultivated biomass. The total lipid content and productivity, and the biodiesel yield and productivity are represented in Fig. 6. The result demonstrated that the lipid content of the biomass increased with increasing $CO_2$ concentration. The total lipid content obtained for 10, 15, 20, and 25% $CO_2$ biomass was 47, 50, 53, and 56%, respectively. The lipid productivity obtained for 10, 15, 20, and 25% $CO_2$ was 0.04, 0.03, 0.05, and 0.1 g $L^{-1}$ $h^{-1}$, respectively. The ability of lipid-producing bacteria to synthesize lipids is directly influenced by the availability and concentration of carbon in their growth medium. Carbon functions as a fundamental component of fatty acids, which are the main components of lipids (Mishra et al., 2019). High carbon-to-nitrogen ratios generally favor lipid accumulation. Bacteria exhibit a preference for storing surplus carbon in the form of lipids rather than utilizing it for protein synthesis when carbon is more plentiful than nitrogen (Karemore and Sen, 2015). The biodiesel yield obtained for 10, 15, 20, and 25% $CO_2$ biomass was 74.5, 80, 87.5, and 89.3%, respectively. Similarly, the biodiesel productivity obtained for 10, 15, 20, and 25% $CO_2$ was 0.06, 0.05, 0.09, and 0.2 g $L^{-1}$ $h^{-1}$, respectively. The production of biodiesel commonly involves the utilization of a transesterification process in which lipids undergo a reaction with an alcohol in the presence of a catalyst. This reaction results in the formation of FAMEs. The potential yield of biodiesel per unit of biomass increases proportionally with the lipid content of the biomass (Thangam et al., 2021). Hence, the highest lipid and biodiesel yield is obtained in the case of 25% $CO_2$.

The 3D surface plot in Fig. 6 (b) represents lipid concentration over time and column height based on the lipid accumulation model in a bioreactor. The lipid concentration increases rapidly at early times (0–150 h), particularly near the bottom of the column ($x \approx 0$), indicating rapid lipid biosynthesis. It reaches a plateau after 200 h, suggesting that lipid biosynthesis saturates either due to nutrient depletion, oxygen limitation, or a saturation of biomass capacity. Lipid concentration decreases with increasing height (moving toward the top of the column). This gradient is consistent with biomass and substrate consumption from bottom to top, as biomass likely accumulates more near the bottom (higher substrate availability and longer residence time). Maximum lipid concentration is achieved near the bottom region ($x = 0$) after sufficient time ($t > 200$ h), reaching about 20 g $L^{-1}$. This aligns with the

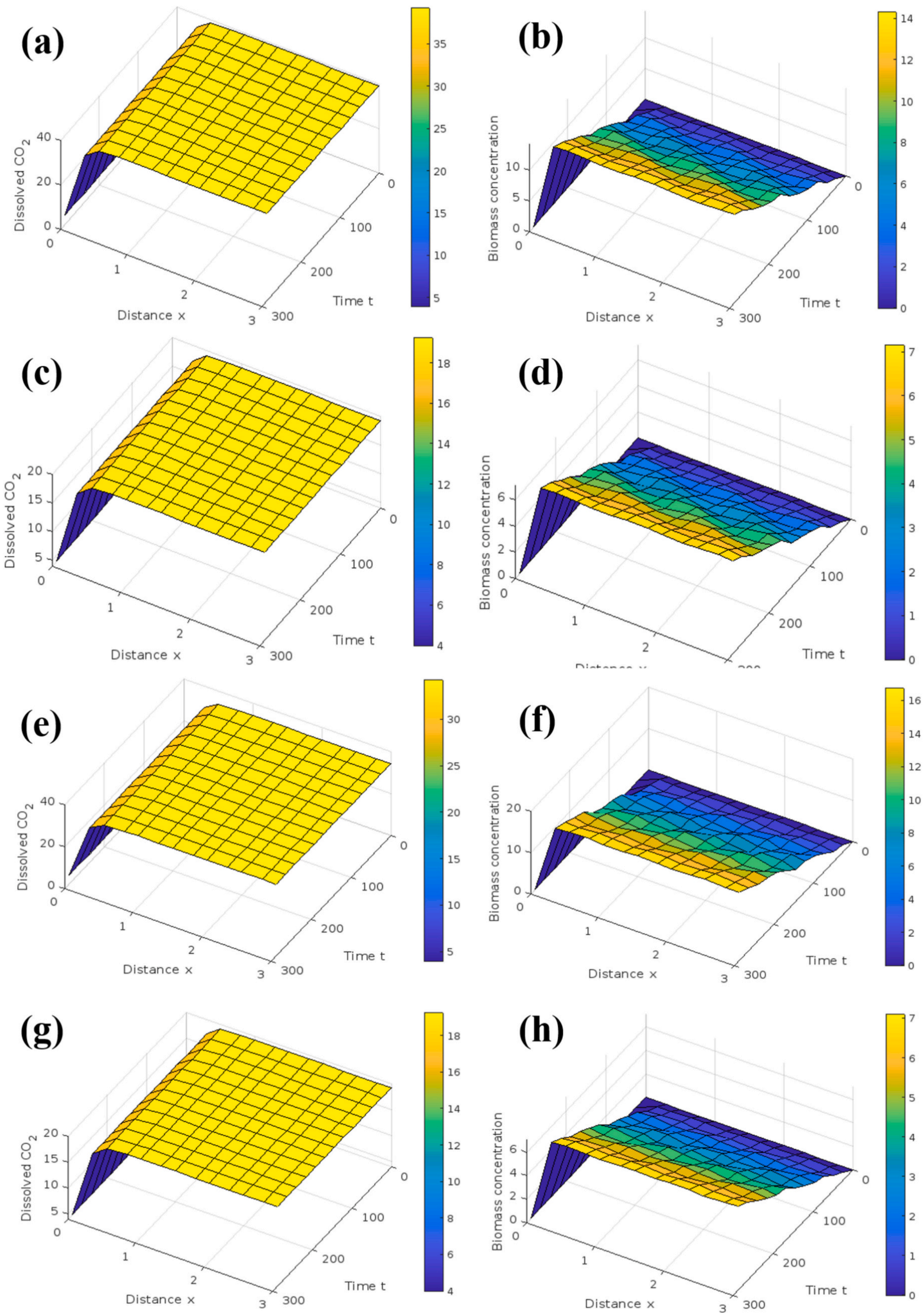


**Fig. 3.** The behavior of a reactor system under different diffusivity and gas flow rate conditions, showing the dissolved $CO_2$ and biomass concentration over time and distance. **(a)** dissolved $CO_2$ and **(b)** biomass correlation when diffusivity is increased by 50%; **(c)** dissolved $CO_2$ and **(d)** biomass correlation when diffusivity is decreased by 50%; **(e)** dissolved $CO_2$ and **(f)** biomass correlation when gas flow rate is increased by 50%; **(g)** dissolved $CO_2$ and **(h)** biomass correlation when gas flow rate is decreased by 50%.

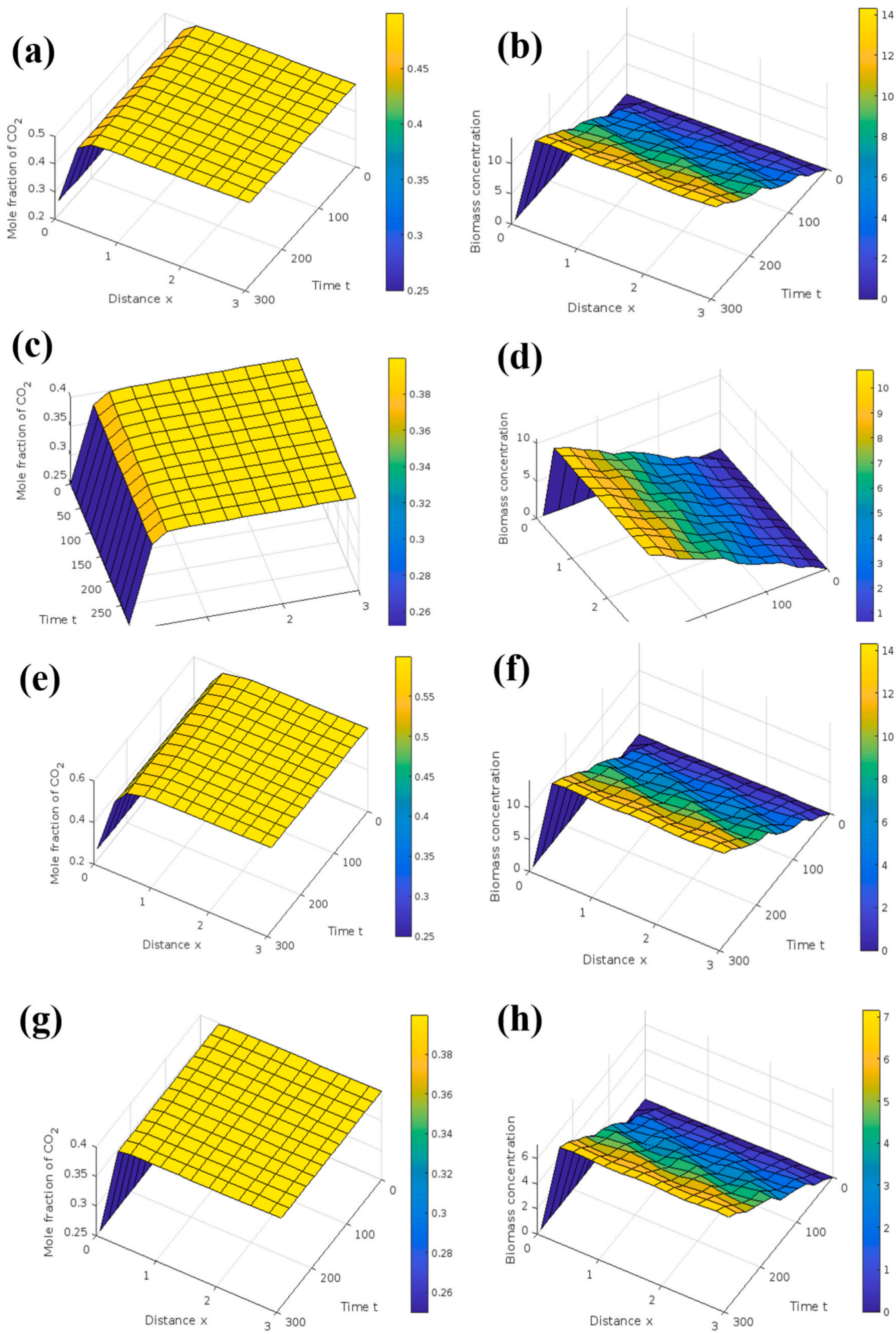


**Fig. 4.** The behavior of a reactor system under different temperature and pressure conditions, showing the dissolved $CO_2$ and biomass concentration over time and distance. **(a)** dissolved $CO_2$ and **(b)** biomass correlation when temperature is increased by 50%; **(c)** dissolved $CO_2$ and **(d)** biomass correlation when temperature is decreased by 50%; **(e)** dissolved $CO_2$ and **(f)** biomass correlation when pressure is increased by 50%; **(g)** dissolved $CO_2$ and **(h)** biomass correlation when pressure is decreased by 50%.

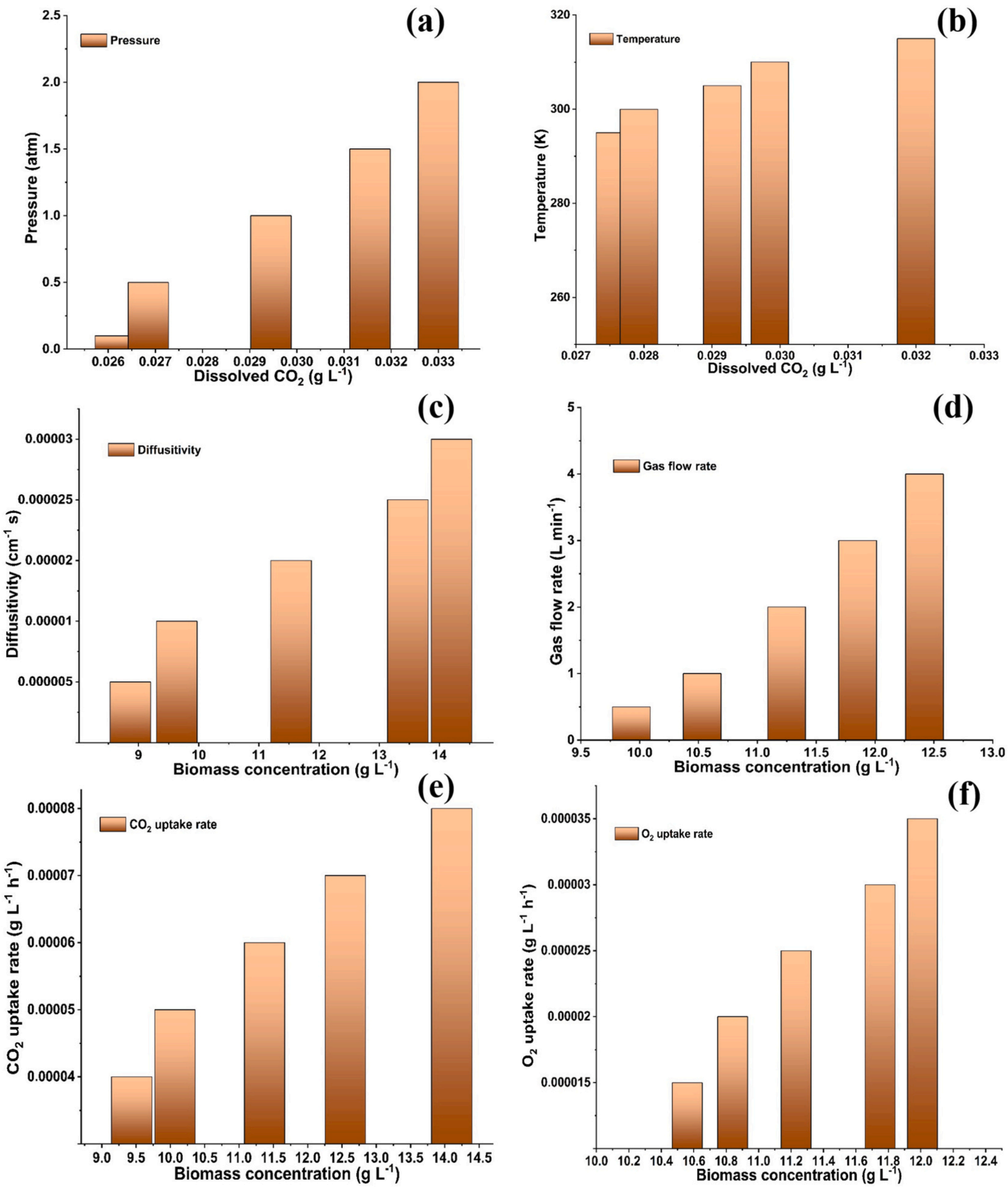


**Fig. 5.** Variation of different physical and biochemical parameters as functions of dissolved $CO_2$ and biomass concentration in the reactor system: **(a)** pressure, **(b)** temperature, **(c)** diffusivity, **(d)** gas flow rate, **(e)** $CO_2$ uptake rate, and **(f)** $O_2$ uptake rate.

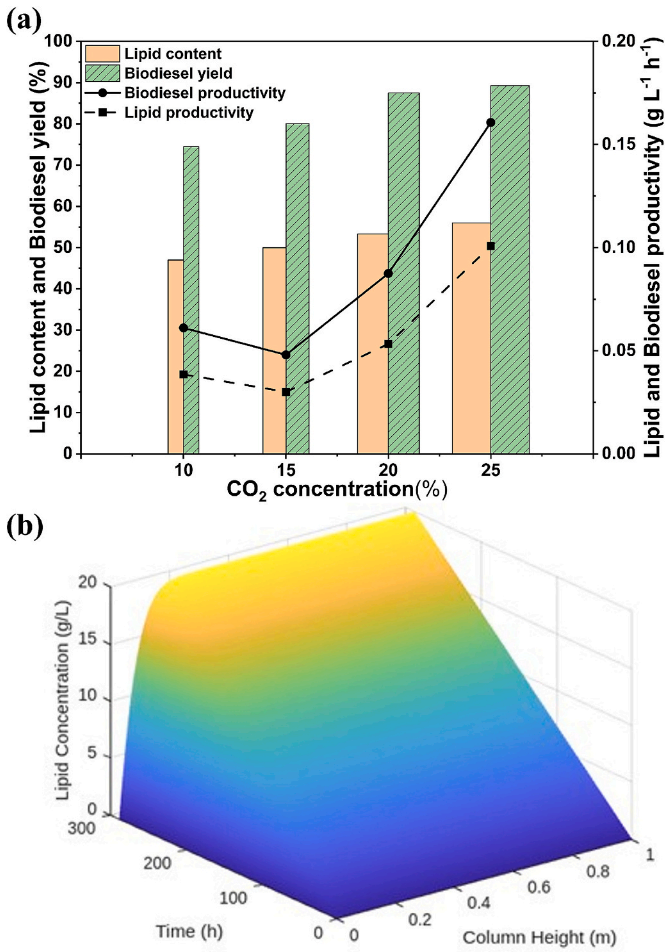


**Fig. 6.** **(a)** The lipid content, biodiesel yield, and lipid and biodiesel productivity obtained during the experimental studies, and **(b)** Estimation of lipid production in the bioreactor.

model assumption that lipid accumulation is a function of local biomass, substrate, and oxygen, all more favorable at lower heights and early stages. The flat yellow region indicates the system has reached steady-state lipid production in the lower zones. This could reflect that the microbial cells have reached lipid storage capacity, and the rate of production equals degradation or extraction. The result implies that lipid production in the bioreactor is time and spatially dependent, optimizing aeration, mixing, or nutrient feeding could push the lipid yield higher or extend the productive phase (Mishra et al., 2019).

Supplementary Table S1 displays the FAME composition derived from the transesterification process implemented on the extracted lipids of *B. cereus* SSLMC2 biomass. The FAME primarily consisted of saturated fatty acid (SFA), which majorly consisted of butyric acid, heptadecanoic acid, linolelaidic acid, linoleic acid, arachidic acid, γ-linolenic acid, and linolenic acid. The percentages of monounsaturated fatty acid (MUFA), polyunsaturated fatty acid (MUFA), and unsaturated fatty acid (UFA) were 8.7, 23.1, and 68.2 wt% for 10% $CO_2$; 5.7, 28.0 and 66.2 wt% for 15% $CO_2$; 5.7, 34.2 and 60.1 wt% for 20% $CO_2$; 9.3, 35.1 and 55.6 wt% for 25% CO2, respectively. The ratio of SFA:UFA obtained for the FAME was around 72:68, 74:66, 69:60, and 76:56 for 10, 15, 20 and 25% $CO_2$, respectively. Hence, the obtained composition of the FAME and biodiesel yield is deemed appropriate for the production of biodiesel derived from bacterial biomass. Fatty acids consist of carbon chains that exhibit a range of lengths. Feedstocks characterized by elevated carbon content show the presence of fatty acids possessing long carbon chains (Lindehoff et al., 2024). Transesterification generally leads to increased FAME yields when the carbon-to-oxygen ratios are higher (Zhang et al., 2015). Hence, *B. cereus* SSLMC2 cultivated in an aerobic environment of 20% $O_2$ and an increased 25% $CO_2$ favored the formation of FAMEs. An increased carbon content within the biomass typically correlates with enhanced transesterification efficiency, thereby leading to elevated yields of FAMEs. The reason for this phenomenon is that lipids possessing longer carbon chains exhibit a higher production of FAMEs per unit mass during transesterification (Mishra et al., 2016). Table 2 reports a comparison of different microorganisms used and their efficiency for biodiesel production.

### 3.5. Biodiesel quality assessment

The quality of biofuel can be determined through empirical analysis of its fatty acid composition. The efficacy of utilizing a set of equations for the determination of biodiesel properties has been demonstrated by many researchers (Nayak et al., 2016; Tanimura et al., 2014). This study involves the calculation and comparison of the essential characteristics of biofuel derived from *B. cereus* SSLMC2 biomass, with the biodiesel properties reported in international American Society for Testing and Materials (ASTM D6751) standards and biodiesel produced from microalgae and yeast, as well as conventional petro-diesel (Table 3). The cetane number measures fuel ignition in a diesel engine's combustion chamber. Higher cetane fuels ignite faster and burn more efficiently. The saponification value is used to estimate free fatty acids (FFAs) in lipids. FFAs can cause soap formation, equipment fouling, lower biodiesel yield, and quality issues (Lin, 2022). Biodiesel with better oxidative stability and cold-flow properties is made from lipids with lower iodine values, indicating a lower degree of unsaturation. Oxidation-induced degradation during storage and use is reduced in biodiesel with higher oxidative stability, improving fuel stability, engine deposits, combustion efficiency, and emissions. Burning biodiesel with higher HV values produces more heat energy per unit mass, improving fuel efficiency (Arora et al., 2017).

The biodiesel obtained in this study exhibits a strong correlation between its specific gravity and viscosity values and the ASTM standards. The utilization of biodiesel with a lower cloud point mitigates the potential hazards of fuel system blockages and engine malfunctions that arise from the solidification of fuel caused by cold weather conditions. The long-chain saturated factor pertains to the relative abundance of long-chain saturated fatty acids within the biodiesel feedstock. This characteristic has the potential to impact fuel properties, oxidative stability, cold flow properties, and engine performance (Barla et al., 2022). The CFP of biodiesel is a crucial parameter that indicates the lowest temperature at which the fuel can flow through a standardized filtration apparatus without clogging (Nayak et al., 2016). Hence, the cultivation of *B. cereus* SSLMC2 in various $CO_2$ concentrations yields biodiesel properties that demonstrate its potential to replace petroleum-based fossil fuels, as it fulfills the ASTM standards.

## 4. Future scope and recommendations

Scaling up the bioreactor system to an industrial scale can enhance the research by facilitating a comprehensive understanding of the system's efficiency in actual industrial settings. Potential future research could explore alternative chemolithotrophs or microbial consortia to improve the effectiveness of $CO_2$ mitigation. A pilot study conducted in diverse industrial settings could offer valuable insights into the adaptability and efficacy of the bioreactor system in various environments. Subsequent investigations may focus on enhancing the efficiency of recovering and refining bio-products, including bio-oil, bio-fertilizer, bioethanol, and bio-polymers, from the collected biomass. Alliances with industries and technology developers should be formed to streamline the shift from research to practical implementation. Design and implement procedures for ongoing surveillance of the bioreactor system to guarantee its sustained effectiveness and reliability. Real-time data analysis can facilitate the early detection of problems, so assuring

**Table 2**

A comparison of the biomass and lipid productivity, as well as the biodiesel yield, obtained from different microorganisms using different carbon sources.

| Microorganism | Carbon source | Biomass productivity (g $L^{-1}$ $d^{-1}$) | Lipid productivity (g $L^{-1}$ $d^{-1}$) | Biodiesel yield (%) | Reference |
|---|---|---|---|---|---|
| *Scenedesmus* sp. | 2.5% $CO_2$ | 0.3 | 0.18 | 80.2 | (Nayak et al., 2018) |
| *Micractinium* sp. ME05 | 10% vinasse | 0.32 | 3.4 | 74 | (Engin et al., 2018) |
| *Serratia* sp. ISTD04 | 25 g $L^{-1}$ municipal solid sludge +50 mM $NaHCO_3$ | 5.06 | 3.07 | 11.21 | (Kumar and Thakur, 2018) |
| DS-7 | Lactose | 0.86 | 0.73 | 38.18 | (Ranjan et al., 2019) |
| *Phormidium valderianum* BDU 20041 | 15% $CO_2$ | 0.03 | 12.74% | 58.16 | (Dineshbabu et al., 2017) |
| *B. cereus* SSLMC2 | 10% $CO_2$ | 0.04 | 0.04 | 74.47 | Present study |
| *B. cereus* SSLMC2 | 15% $CO_2$ | 0.04 | 0.03 | 80.0 | Present study |
| *B. cereus* SSLMC2 | 20% $CO_2$ | 0.03 | 0.05 | 87.50 | Present study |
| *B. cereus* SSLMC2 | 25% $CO_2$ | 0.05 | 0.10 | 89.29 | Present study |

**Table 3**

Biodiesel property assessment and comparison with reported microalgal, yeast-derived biodiesel, petro-diesel, and ASTM standards.

| Biodiesel properties | Present study (% $CO_2$) | | | | Microalgal biodiesel (Nayak et al., 2018) | Yeast biodiesel (Tanimura et al., 2014) | Petro-diesel (Kumar and Thakur, 2018) | ASTM standards (McCormick and Moriarty, 2023) |
|---|---|---|---|---|---|---|---|---|
| | 10% | 15% | 20% | 25% | | | | |
| Cetane number | 50.1 | 52.13 | 54.12 | 55.21 | 56.52 | 54.3–55.2 | 40–55 | ≥47 |
| Saponification value | 0.24 | 0.22 | 0.25 | 0.20 | 0.21 | 0.22–0.27 | 0.002 | ≤0.50 |
| Degree of unsaturation | 83.21 | 83.87 | 84.56 | 85.10 | 83.14 | 84.11 | – | – |
| Higher heating value | 40.20 | 40.35 | 40.65 | 40.87 | 40.20 | 40.5–40.7 | 43 | 37.6–40.5 |
| Cloud point | 6.54 | 6.87 | 7.10 | 7.45 | 7.35 | 2.9–4.6 | −5 | ≤ −10 |
| Specific gravity | 0.85 | 0.84 | 0.84 | 0.86 | 0.87 | 0.87 | 0.88 | 0.85 |
| Viscosity | 4.14 | 4.15 | 4.14 | 4.16 | 4.61 | 4.4 | 3.4 | 1.9–6 |
| Long chain saturated factor | 4.56 | 4.45 | 4.98 | 4.65 | – | 4.87 | – | – |
| Cold filter plugging point | −2.5 | −2.3 | −2.4 | −1.8 | – | −1.40 | −6 | – |
| Oxidative stability | 1 | 1 | 1 | 1 | – | – | 1 | 3 |

the achievement of optimal performance. Promote additional investment in research and development to investigate novel technologies that may improve the effectiveness of $CO_2$ capture and conversion methodologies. Ongoing innovative efforts are crucial to establish bio-mitigation as a feasible and economically efficient approach for large-scale industrial applications.

## 5. Conclusions

This study demonstrates the feasibility of integrating microbial $CO_2$ utilization with mathematical modeling to evaluate biomass production and biodiesel potential in a bubble column bioreactor using *Bacillus cereus* SSLMC2. The developed model successfully captured the coupled interactions between bacterial growth kinetics, $CO_2$ uptake, and gas–liquid mass transfer dynamics within the reactor system. Model simulations showed good agreement with experimental observations, confirming the reliability of the modeling framework in predicting biomass growth and $CO_2$ bio-mitigation under varying operating conditions. The results highlight the ability of *B. cereus* SSLMC2 to utilize $CO_2$ as a carbon source, resulting in enhanced biomass accumulation and lipid production suitable for biodiesel synthesis. The highest lipid and biodiesel productivity of 0.1 and 0.2 g $L^{-1}$ $h^{-1}$, respectively, was also achieved for 25% $CO_2$. Furthermore, the study demonstrates how operational parameters such as gas flow rate and dissolved gas concentrations influence reactor performance, providing critical insights for optimizing $CO_2$ conversion efficiency and microbial productivity. A key contribution of this work is the development of an integrated modeling approach that links microbial metabolism with transport phenomena in bubble column reactors. This framework provides a valuable predictive tool for optimizing reactor design and operation while reducing experimental costs and time. Overall, the findings support the potential of bacterial systems as sustainable platforms for simultaneous $CO_2$ mitigation and biofuel production. Future studies should focus on scale-up validation and integration of metabolic pathway modeling to further enhance process efficiency for industrial applications.

## Abbreviations

| | | |
|---|---|---|
| L | liquid phase | – |
| G | gaseous phase | – |
| (x,t) | differential section of the column | – |
| $C_L$ | $O_2$ concentration in liquid phase | $ML^{-3}$ |
| $D_L$ | diffusivity of $O_2$ & $CO_2$ in the liquid phase | $L^2T^{-1}$ |
| $u_L$ | superficial liquid velocity | $LT^{-1}$ |
| $R_o$ | $O_2$ uptake rate | $ML^{-3}T^{-1}$ |
| X | biomass concentration | $ML^{-3}$ |
| S | substrate concentration | $ML^{-3}$ |
| C | $O_2$ concentration | $ML^{-3}$ |
| $k_L$ | mass transfer coefficient | $LT^{-1}$ |
| a | interfacial area | $L^2$ |
| $C_L^*$ | equilibrium $O_2$ concentration | $ML^{-3}$ |
| p | pressure | $ML^{-1}T^{-2}$ |
| $C_G$ | mole fraction of $O_2$ in the gas phase | - |
| $D_G$ | diffusivity of $O_2$ in the gas phase | $L^2T^{-1}$ |
| $u_G$ | superficial gas velocity | $LT^{-1}$ |
| R | gas constant | $L^2T^{-2}K^{-1}$ |
| T | temperature | K |
| $M_{O2}$ | molecular mass | – |
| $\varepsilon_G$ | gas hold up | – |
| $\varepsilon_L$ | liquid hold up | – |
| $C_{Go}$ | initial mole fraction of $O_2$ in the gas phase | $ML^{-3}$ |
| $C_{Lo}$ | initial $O_2$ concentration in liquid phase | $ML^{-3}$ |
| h | height of the column | L |
| $S_L$ | $CO_2$ concentration in liquid phase | $ML^{-3}$ |
| $S_G$ | mole fraction of $CO_2$ in the gas phase | – |
| $R_S$ | $CO_2$ uptake rate | $ML^{-3}T^{-1}$ |
| $M_{CO2}$ | molecular mass | – |
| $S_L^*$ | equilibrium $CO_2$ concentration | $ML^{-3}$ |
| $X_L$ | biomass concentration in liquid phase | $ML^{-3}$ |
| $R_X$ | Cell growth rate | $ML^{-3}T^{-1}$ |

## CRediT authorship contribution statement

**Rachael Jovita Barla:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Suresh Gupta:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization. **Smita Raghuvanshi:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biteb.2026.102779.

## Data availability

Data will be made available on request.

## References


Abu-Reesh, I.M., 2025. Mathematical model-based optimization of continuous flow photobioreactor operating at steady state using MATLAB optimization function. Arab. J. Sci. Eng. 50, 3779–3792. https://doi.org/10.1007/s13369-024-09155-7.

Al Ketife, A.M.D., Judd, S., Znad, H., 2016. A mathematical model for carbon fixation and nutrient removal by an algal photobioreactor. Chem. Eng. Sci. 153, 354–362. https://doi.org/10.1016/j.ces.2016.07.042.

Alshalif, A.F., Irwan, J.M., Othman, N., Al-Gheethi, A.A., Shamsudin, S., 2022. A systematic review on bio-sequestration of carbon dioxide in bio-concrete systems: a future direction. Eur. J. Environ. Civ. Eng. 26 (3), 1209–1228. https://doi.org/10.1080/19648189.2020.1713899.

Anand, A., Raghuvanshi, S., Gupta, S., 2020. Trends in carbon dioxide ($CO_2$) fixation by microbial cultivations. Curr. Sustain. Renew. Energy Rep. 7 (2), 40–47. https://doi.org/10.1007/s40518-020-00149-1.

Anand, A., Raghuvanshi, S., Gupta, S., 2023. Sustainable approach for simultaneously reducing - $CO_2$ and NO emissions from synthetic industrial flue gases using bacterial consortium and domestic wastewater in a suspended glass bioreactor. Biomass Conv. Bioref. 14, 11911–11926. https://doi.org/10.1007/s13399-023-03934-2.

Anand, A., Raghuvanshi, S., Gupta, S., 2024. Assessing the bacterial consortium's potential to bio-mitigate $CO_2$ and $SO_2$ from simulated flue gas, wastewater bioremediation, and product characterization. Process Biochem. 147, 86–100. https://doi.org/10.1016/j.procbio.2024.08.003.

Anjos, M., Fernandes, B.D., Vicente, A.A., Teixeira, J.A., Dragone, G., 2013. Optimization of $CO_2$ bio-mitigation by Chlorella vulgaris. Bioresour. Technol. 139, 149–154. https://doi.org/10.1016/j.biortech.2013.04.032.

Arora, N., Gulati, K., Patel, A., Pruthi, P.A., Poluri, K.M., Pruthi, V., 2017. A hybrid approach integrating arsenic detoxification with biodiesel production using oleaginous microalgae. Algal Res. 24, 29–39. https://doi.org/10.1016/j.algal.2017.03.012.

Barla, R.J., Raghuvanshi, S., Gupta, S., 2022. Process integration for the biodiesel production from biomitigation of flue gases. In: Singh, B., Guldhe, A. (Eds.), Waste and Biodiesel. Elsevier Ltd, pp. 191–215. https://doi.org/10.1016/B978-0-12-823958-2.00007-0.

Barla, R.J., Gupta, S., Raghuvanshi, S., 2024a. Sustainable synergistic approach to chemolithotrophs—supported bioremediation of wastewater and flue gas. Sci. Rep. 14 (1), 1–25. https://doi.org/10.1038/s41598-024-67053-2.

Barla, R.J., Raghuvanshi, S., Gupta, S., 2024b. Reforming $CO_2$ bio-mitigation utilizing Bacillus cereus from hypersaline realms in pilot-scale bubble column bioreactor. Sci. Rep. 14, 6354. https://doi.org/10.1038/s41598-024-56965-8.

Bok, F., Moog, H.C., Brendler, V., 2023. The solubility of oxygen in water and saline solutions. Front. Nucl. Eng. 2, 1158109. https://doi.org/10.3389/fnuen.2023.1158109.

Dineshbabu, G., Uma, V.S., Mathimani, T., Deviram, G., Arul Ananth, D., Prabaharan, D., Uma, L., 2017. On-site concurrent carbon dioxide sequestration from flue gas and calcite formation in ossein effluent by a marine cyanobacterium Phormidium valderianum BDU 20041. Energy Convers. Manag. 141, 315–324. https://doi.org/10.1016/j.enconman.2016.09.040.

Doucha, J., Straka, F., Lívanský, K., 2005. Utilization of flue gas for cultivation of microalgae (Chlorella sp.) in an outdoor open thin-layer photobioreactor. J. Appl. Phycol. 17 (5), 403–412. https://doi.org/10.1007/s10811-005-8701-7.

Dunn, I.J., Heinzle, E., Ingham, J., Prenosil, J.E., 2013. Biological Reaction Engineering: Dynamic Modelling Fundamentals with Simulation Examples, Third ed. Wiley-VCH.

Eloka-Eboka, A.C., Inambao, F.L., 2017. Effects of $CO_2$ sequestration on lipid and biomass productivity in microalgal biomass production. Appl. Energy 195, 1100–1111. https://doi.org/10.1016/j.apenergy.2017.03.071.

Emerson, D.F., Woolston, B.M., Liu, N., Donnelly, M., Currie, D.H., Stephanopoulos, G., 2019. Enhancing hydrogen-dependent growth of and carbon dioxide fixation by Clostridium ljungdahlii through nitrate supplementation. Biotechnol. Bioeng. 116 (2), 294–306. https://doi.org/10.1002/bit.26847.

Engin, I.K., Cekmecelioglu, D., Yücel, A.M., Oktem, H.A., 2018. Evaluation of heterotrophic and mixotrophic cultivation of novel Micractinium sp. ME05 on vinasse and its scale up for biodiesel production. Bioresour. Technol. 251, 128–134. https://doi.org/10.1016/j.biortech.2017.12.023.

Ganta, U., Rai, A., Singh, S., Chakrabarty, J., Ghanta, K.C., Dutta, S., 2019. Biomitigation of $CO_2$ and Extraction of Biomolecules using Leptolyngbya sp. J. Environ. Eng. 145 (6), 2536–2548. https://doi.org/10.1061/(asce)ee.1943-7870.0001535.

Gaur, A., 2015. Isolation and characterization of halotolerant *Bacillus* sp with extra cellular A-Amylase production potential from Sambhar Salt Lake, India. Int. Res. J. Environ. Sci. 4 (6), 2319–2329.

Gevantman, L.H., 2015. Solubility of selected gases in water. In: Gevantman, L.H. (Ed.), CRC Handbook of Chemistry and Physics, 97th ed. CRC Press, pp. 5–8.

Karemore, A., Sen, R., 2015. Green integrated process for mitigation of municipal and industrial liquid and solid waste mixes for enhanced microalgal biomass and lipid synthesis for biodiesel. RSC Adv. 5 (87), 70929–70938. https://doi.org/10.1039/c5ra11670h.

Kumar, M., Thakur, I.S., 2018. Municipal secondary sludge as carbon source for production and characterization of biodiesel from oleaginous bacteria. Bioresour. Technol. Rep. 4, 106–113. https://doi.org/10.1016/j.biteb.2018.09.011.

Lin, C., 2022. Determination of Cetane Number from Fatty Acid Compositions and Structures of biodiesel. Processes 10 (8), 1502. https://doi.org/10.3390/pr10081502.

Lindehoff, E., Mattsson, L., Olofsson, M., Svensson, F., Farnelid, H., Legrand, C., 2024. Biomass performance and stability of 5-year outdoor microalgal cultivation for $CO_2$ removal from cement flue gas. Bioresour. Technol. Rep. 25, 101730. https://doi.org/10.1016/j.biteb.2023.101730.

McCormick, R., Moriarty, K., 2023. Biodiesel Handling and Use Guide: Sixth Edition. www.nrel.gov/publications.

Mishra, S., Gupta, S., Raghuvanshi, S., Pal, P., 2016. Energetic assessment of fixation of $CO_2$ and subsequent biofuel production using B. Cereus SM1 isolated from sewage treatment plant. Bioprocess Biosyst. Eng. 39 (8), 1247–1258. https://doi.org/10.1007/s00449-016-1603-z.

Mishra, S., Chauhan, P., Gupta, S., Raghuvanshi, S., Singh, R.P., Jha, P.N., 2017. $CO_2$ sequestration potential of halo-tolerant bacterium Pseudomonas aeruginosa SSL-4 and its application for recovery of fatty alcohols. Process. Saf. Environ. Prot. 111, 582–591. https://doi.org/10.1016/j.psep.2017.08.013.

Mishra, S., Pahari, S., Siva, K., Mohanty, S., Gupta, S., Raghuvanshi, S., 2018. Investigation on $CO_2$ bio-mitigation using Halomonas stevensii in laboratory scale bioreactor: Design of downstream process and its economic feasibility analysis. J. $CO_2$ Util. 24, 274–286. https://doi.org/10.1016/j.jcou.2018.01.018.

Mishra, S., Raghuvanshi, S., Gupta, S., 2019. Carbon dioxide to bio-fuels by mixed and pure microbial cultures isolated from activated sludge: Relative evaluation of $CO_2$ fixation, biodiesel production, and thermodynamic analysis. Greenh. Gases Sci. Technol. 9 (6), 1135–1157. https://doi.org/10.1002/ghg.1922.

Morchain, J., 2017. Tools for Bioreactor Modeling and simulation. In: Morchain, J. (Ed.), Bioreactor Modeling, First ed. ISTE Press - Elsevier, pp. 1–27. https://doi.org/10.1016/b978-1-78548-116-1.50001-x.

Nayak, M., Karemore, A., Sen, R., 2016. Sustainable valorization of flue gas $CO_2$ and wastewater for the production of microalgal biomass as a biofuel feedstock in closed and open reactor systems. RSC Adv. 6 (94), 9111–91120. https://doi.org/10.1039/c6ra17899e.

Nayak, M., Dhanarajan, G., Dineshkumar, R., Sen, R., 2018. Artificial intelligence driven process optimization for cleaner production of biomass with co-valorization of wastewater and flue gas in an algal biorefinery. J. Clean. Prod. 201, 1092–1100. https://doi.org/10.1016/j.jclepro.2018.08.048.

Onyeaka, H., Ekwebelem, O.C., 2023. A review of recent advances in engineering bacteria for enhanced $CO_2$ capture and utilization. Int. J. Environ. Sci. Technol. 20 (4), 4635–4648. https://doi.org/10.1007/s13762-022-04303-8.

Paleari, S., 2024. The EU policy on climate change, biodiversity and circular economy: moving towards a Nexus approach. Environ. Sci. Pol. 151, 103603. https://doi.org/10.1016/j.envsci.2023.103603.

Panda, T., 2011. Bioreactor Modeling. In: Panda, T. (Ed.), Bioreactors: Analysis and Design, First ed. McGraw-Hill Education, pp. 364–387 https://www.accessengineeringlibrary.com/content/book/9780070704244/chapter/chapter6.

Pekkoh, J., Duangjan, K., Phinyo, K., Kaewkod, T., Ruangrit, K., Thurakit, T., Pumas, C., Pathom-aree, W., Cheirsilp, B., Gu, W., Wang, G., Chaichana, C., Srinuanpan, S., 2023. Turning waste $CO_2$ into value-added biorefinery CO-products using cyanobacterium Leptolyngbya sp. KC45 as a highly efficient living photocatalyst. Chem. Eng. J. 460, 141765. https://doi.org/10.1016/j.cej.2023.141765.

Ranjan, A., Dutta, K., Verma, P., Daverey, A., 2019. High lipid accumulating bacteria isolated from dairy effluent scum grown on dairy wastewater as potential biodiesel feedstock. J. Environ. Manag. 252, 109686. https://doi.org/10.1016/j.jenvman.2019.109686.

Rittmann, B.E., McCarty, P.L., 2001. Energy Reactions. In: Environmental Biotechnology: Principles and Applications, First Edit. McGraw-Hill Education. https://www.accessengineeringlibrary.com/content/book/9781260440591/toc-chapter/chapter2/section/section5.

Rodriguez-Jara, M., Ramírez-Castelan, C.E., Samano-Perfecto, Q., Ricardez-Sandoval, L. A., Puebla, H., 2023. Robust control designs for microalgae cultivation in continuous photobioreactors. Int. J. Chem. React. Eng. 21 (4), 521–535. https://doi.org/10.1515/ijcre-2022-0115.

Setia, R., Marschner, P., Baldock, J., Chittleborough, D., Verma, V., 2011. Relationships between carbon dioxide emission and soil properties in salt-affected landscapes. Soil Biol. Biochem. 43 (3), 667–674. https://doi.org/10.1016/j.soilbio.2010.12.004.

Schügerl, K., 1982. Some basic principles for the layout of tower bioreactors. J. Chem. Tech. Biotechnol. https://doi.org/10.1002/jctb.5030320111.

Shuler, M.L., Kargi, F., Delisa, M., 2017. How Cells Grow. In: Shuler, M.L., Kargi, F., Delisa, M. (Eds.), Bioprocess Engineering: Basic Concepts. Pearson, pp. 165–200.

Tanimura, A., Takashima, M., Sugita, T., Endoh, R., Kikukawa, M., Yamaguchi, S., Sakuradani, E., Ogawa, J., Shima, J., 2014. Selection of oleaginous yeasts with high lipid productivity for practical biodiesel production. Bioresour. Technol. 153, 230–235. https://doi.org/10.1016/j.biortech.2013.11.086.

Tebbani, S., Lopes, F., Filali, R., Dumur, D., Pareau, D., 2014. Nonlinear predictive control for maximization of $CO_2$ bio-fixation by microalgae in a photobioreactor. Bioprocess Biosyst. Eng. 37 (1), 83–97. https://doi.org/10.1007/s00449-013-0928-0.

Thangam, K.R., Santhiya, A., Sri, S.R.A., Mubarakali, D., Karthikumar, S., Kumar, R.S., Thajuddin, N., Rahul, M., Varalakshmi, P., Moorthy, I.G., Pugazhendhi, A., 2021. Bio-refinery approaches based concomitant microalgal biofuel production and wastewater treatment. Sci. Total Environ. 785, 147267. https://doi.org/10.1016/j.scitotenv.2021.147267.

Trapé, D.V., López, O.V., Villar, M.A., 2024. Exploitation of Biomass to the Integrated production of Bioethanol and Poly(hydroxyalkanoate)s. Bioenergy Res. 17 (1), 27–41. https://doi.org/10.1007/s12155-023-10661-z.

UN DESA, 2023. SDGs Report 2023. In: the Sustainable development Goals Report 2023: special Edition, p. 80. https://unstats.un.org/sdgs/report/2023/.

Viswanaathan, S., Perumal, P.K., Sundaram, S., 2022. Integrated approach for carbon sequestration and wastewater treatment using algal–bacterial consortia: Opportunities and challenges. Sustainability (Switzerland) 14 (3), 1075. https://doi.org/10.3390/su14031075.

Yadav, G., Dash, S.K., Sen, R., 2019. A biorefinery for valorization of industrial waste-water and flue gas by microalgae for waste mitigation, carbon-dioxide sequestration and algal biomass production. Sci. Total Environ. 688, 129–135. https://doi.org/10.1016/j.scitotenv.2019.06.024.

Zhang, Y., Li, Y., Zhang, X., Tan, T., 2015. Biodiesel production by direct transesterification of microalgal biomass with co-solvent. Bioresour. Technol. 196, 712–715. https://doi.org/10.1016/j.biortech.2015.07.052.

Zhou, X., Yi, H., Tang, X., Deng, H., Liu, H., 2012. Thermodynamics for the adsorption of $SO_2$, NO and $CO_2$ from flue gas on activated carbon fiber. Chem. Eng. J. 200–202 (2), 399–404. https://doi.org/10.1016/j.cej.2012.06.013.